Advanced C, Joystick Interfacing

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From: ajmyrvold@violet.waterloo.edu (Alan Myrvold)
Newsgroups: comp.sys.ibm.pc
Subject: Joystick routines
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Keywords: Game Port, Joy Stick
How to read the joystick on an IBM PC.

This program, written for Turbo C, version 1.0 shows how the joystick
postion may be read. I am indebted to uunet!netxcom!jallen (John Allen) for
answering my previous posting on the subject and providing technical
assistance.
Text from his EMAIL to me is included in the program.
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----------------------------- JOY.C    ----------------------------
/* Demonstration of reading joy stick postition
    Written March 24, 1988
    Written by : ajmyrvold@violet.waterloo.edu (Alan J. Myrvold)
Technical assistance from : uunet!netxcom!jallen (John Allen)
Turbo C Version 1.0
*/

#include <stdio.h>
#include <dos.h>
#include <bios.h>

typedef struct {
    int sw1,sw2;
    int x,y;
    int cenx,ceny;
    } joy_stick;

joy_stick joy;

#define keypressed (bioskey(1) != 0)
#define kb_clear() while keypressed bioskey(0);

void GotoXY(int x,int y)
{
    union REGS r;
    /* Set XY position */
    r.h.ah = 2;
    r.h.bh = 0;            /* Assume Video Page 0 */
    r.h.dh = (char) y;
    r.h.dl = (char) x;
    int86(16,&r,&r);
}

void ClrScr()
{
    union REGS r;

    /* Get video mode */
    r.h.ah = 15;
    int86(16,&r,&r);
```

```
    /* Set video mode */
    r.h.ah = 0;
    int86(16,&r,&r);
}


/*
From: uunet!netxcom!jallen (John Allen)
1.  Trigger the joystick oneshots with an 'out' to 0x201.
    This will set all of the joystick bits on.

2.  Read (in) 0x201, finding:

        Bit             Contents
        0               Joystick A X coordinate
        1               Joystick A Y coordinate
        2               Joystick B X coordinate
        3               Joystick B Y coordinate
        4               Button A 1
        5               Button A 2
        6               Button B 1
        7               Button B 2

3.  Continue reading 0x201 until all oneshots return to zero,
    recording the loop during which each bit falls to zero.
    The duration of the pulse from each oneshot may be used to
    determine the resistive load (from 0 to 100K) from each
    Joystick, as: Time = 24.2msec. + .011 (r) msec.

4.  To do this correctly, I recommend calibrating the joystick;
    have the user move the stick to each corner, then center it,
    while recording the resulting values.

*/

void disp_stick(int line,joy_stick *joy)
{
    GotoXY(0,line);
    printf("sw1 %d sw2 %d",joy -> sw1,joy -> sw2);
    GotoXY(0,line+1);
    printf("x %4d y %4d",joy -> x,joy -> y);
}

void read_stick(int stick,joy_stick *joy)
{
    int k,jx,jy;
    int c,m1,m2,m3,m4,m5;

    /* Define masks for the chosen joystick */
    if (stick == 1) m4 = 1; else
    if (stick == 2) m4 = 4; else
        printf("Invalid stick %d\n",stick);
    m5 = m4 << 1;
    m1 = m4 << 4;
    m2 = m5 << 4;
    m3 = m4 + m5;

    /* Trigger joystick */
    outportb(0x201,0xff);
    c = inportb(0x201);
    /* Read switch settings */
    joy -> sw1 = (c & m1) == 0;
    joy -> sw2 = (c & m2) == 0;
    /* Get X and Y positions */
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    for (k = 0; (c & m3) != 0; k++) {
        if ((c & m4) != 0) jx = k;
        if ((c & m5) != 0) jy = k;
        c = inportb(0x201);
    }
    joy -> x = jx - (joy -> cenx);
    joy -> y = jy - (joy -> ceny);
}

int choose_stick(joy_stick *joy)
{
    int init_swa,init_swb,swa,swb;
    int c,retval;

    printf("Center joystick and press fire, or press any key\n");
    kb_clear();
    outportb(0x201,0xff);
    c = inportb(0x201);
    init_swa = c & 0x30;
    init_swb = c & 0xc0;
    do {
       outportb(0x201,0xff);
       c = inportb(0x201);
       swa = c & 0x30;
       swb = c & 0xc0;
    } while ((swa == init_swa) && (swb == init_swb) && !keypressed);
        if (swa != init_swa) {
       printf("Joystick 1 selected\n");
       retval = 1;
    } else if (swb != init_swb) {
       printf("Joystick 2 selected\n");
       retval = 2;
    } else {
       printf("Keyboard selected\n");
       kb_clear();
       retval = 0;
    }

    if (retval != 0) { /* Determine Center */
       joy -> cenx = joy -> ceny = 0;
       read_stick(retval,joy);
       joy -> cenx = joy -> x;
       joy -> ceny = joy -> y;
    }
    return(retval);
}

main()
{
   int k;

   k = choose_stick(&joy);
   ClrScr();
   if (k != 0) while (!keypressed) {
        read_stick(k,&joy);
        disp_stick(0,&joy);
   }
}
```

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